



Spatio-temporal analysis for obstacle detection in agricultural videos



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ABSTRACT

Autonomous mobile vehicles are becoming commoner in outdoor scenarios for agricultural applications. They must be equipped with a robot navigation system for sensing, mapping, localization, path planning, and obstacle avoidance. In autonomous vehicles, safety becomes a major challenge where unexpected obstacles in the working area must be conveniently addressed. Of particular interest are, people or animals crossing in front of the vehicle or fixed/moving uncatalogued elements in specific positions. Detection of unexpected obstacles or elements on video sequences acquired with a machine vision system on-board a tractor moving in cornfields makes the main contribution to this research. We propose a new strategy for automatic video analysis to detect static/dynamic obstacles in agricultural environments via spatial-temporal analysis. At a first stage obstacles are detected by using spatial information based on spectral colour analysis and texture data. At a second stage temporal information is used to detect moving objects/obstacles at the scene, which is of particular interest in camouflaged elements within the environment. A main feature of our method is that it does not require any training process. Another feature of our approach consists in the spatial analysis to obtain an initial segmentation of interesting objects; afterwards, temporal information is used for discriminating between moving and static objects. To the best of our knowledge in the field of agricultural image analysis, classical approaches make use of either spatial or temporal information, but not both at the same time, making an important contribution. Our method shows favourable results when tested in different outdoor scenarios in agricultural environments, which are really complex, mainly due to the high variability in the illumination conditions, causing undesired effects such as shadows and alternating lighted and dark areas. Dynamic background, camera vibrations and static and dynamic objects are also factors complicating the situation. The results are comparable to those obtained with other state-of-art techniques reported in literature.

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1. Introduction

The concept of fully autonomous vehicles includes a navigation system capable of solving problems such as: mapping, localization, path following, and obstacle avoidance. In recent years, agricultural vehicles (tractors, combines, sprayers, spreaders) have been introduced in agricultural environments to accomplish different tasks including planting, spraying, fertilizing, cultivating, harvesting, thinning, weeding, or inspection. In order to be useful, such vehicles should be equipped with vision-based sensors, which provide the required information to develop these tasks with a good performance.

As mentioned above, autonomous agricultural vehicles need a means of detecting obstructions in their path to avoid

collisions. If the system detects people, animals, other vehicles, or any obstruction, inside the haulage area during autonomous operation, the truck must stop immediately for security reasons. It is therefore important to emphasise that in agricultural applications few systems include an object detection procedure to guarantee safe vehicle navigation. Here comes our initial motivation; we propose an alternative method to detect objects/obstacles from agricultural videos in order to ensure the safety and care of people/animals in the field, but especially those, which are on the tractor trajectory. In addition, not only obstacle detection is of our interest; also, we study obstacles, which are moving in the crop area. Motion analysis is especially valuable for those elements, which can suddenly appear in front of the vehicle.

The videos, used as input data, come from a cornfield during the period of weed removal. They were captured with a unique CCD-based high-resolution camera operating in the visible RGB spectral range. The camera is mounted on board a tractor, which is part of the fleet in the RHEA project [1]; its position is on top of the tractor

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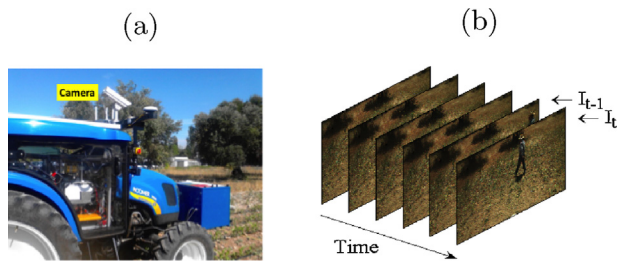


Fig. 1. (a) Camera system arrangement and geometry on top of the tractor. (b) Consecutive frames from an agricultural video.

at a height of 2.20 m and with a pitch angle of 25° with respect to the vertical, Fig. 1a. This arrangement and also the camera calibration are conveniently addressed based on the camera model, and system geometry with the help of landmarks in the cornfield, as described in [2]. Fig. 1b displays six consecutive frames captured with this system. For motion obstacles, four scenarios are identified in our video sequences, they are described in Table 1.

Images from agricultural videos, containing several elements including plants, trees, weeds, soil, objects, and shadows coming from these elements, are complex from the image processing point of view. The complexity increases by the great variability of vehicle and environmental conditions, such as: changing seasons or weather conditions (sunny or cloudy days), time of day, dust or movements produced by the vehicle in movement along the field, and also because of vibrations caused by the tractor engine. As a consequence of these combined effects, a robust obstacle detection system is very demanding.

This paper specifically proposes a particular algorithm to detect dynamic and static elements. To achieve our goal, we have devised an approach based on spatio-temporal analysis. Given an input video, two consecutive video frames are considered for both spatial and temporal analysis. Obstacle segmentation is made by analysing spatial information in the current frame while motion is computed from the first-order spatio-temporal derivatives of two consecutive frames. The spatial and temporal information conveniently combined allows determining objects location and their associated movement if any. One of the major advantages of this representation is that by analysing temporal information, not only obstacle segmentation is improved; it is also possible to identify moving obstacles in front of the tractor trajectory. Motion detection is useful to anticipate and to prevent collisions between autonomous vehicles; this topic is widely discussed in [3]. When the spatial and temporal information is combined, the complexity of motion detection is significantly reduced. Instead of detecting motion with a relatively large number of previous frames, where the scene could change dramatically, we estimate motion based on the first-order temporal difference, but compared against the obstacle segmentation results to determine which of the elements on the scene are really in movement with respect to the tractor motion.

Unlike other reported methods where object detection is exclusively based either on static segmentation or differential methods, such as those based on optical flow computation, we propose a static and dynamic combined approach (SDC) that allows

Table 1

Possible scenarios in our video dataset. Examples of static obstacles: trees, buildings, people standing up, among others. In contrast, moving obstacles can be: people walking, moving tractors, etc.

	Description
Scenario 1	The tractor is stopped and the obstacles are static.
Scenario 2	The tractor is moving and the obstacles are static.
Scenario 3	The tractor is stopped and the obstacles can be static or moving.
Scenario 4	The tractor is moving and the obstacles can be static or moving.

exploiting the advantages of both, making an important contribution in the context of object detection in agricultural environments. The spatial differentiation in optical flow-based methods is assumed by the temporal differentiation, which determines image changes [4]. This combination and the simultaneous separation of spatial and temporal analysis is the main contribution of this proposed approach, which is conveniently compared against existing static and dynamic strategies.

The remainder of the paper is structured as follows. A review of the state-of-the-art is provided in Section 2. Our methodology is extensively explained in Section 3 where the segmentation process is given in Section 3.1, followed by the moving detection in Section 3.2. Performance evaluation of our proposal is presented in Section 4; finally, we conclude the paper in Section 5 where also future trends are provided.

2. Related work

2.1. Obstacle detection

Obstacle avoidance is a classical problem in robotics and in autonomous vehicles; literature on obstacle detection is vast and well known on image processing. Methodologies reported in literature are diverse (see the Refs. [5,6] and the references therein). They include techniques based on: optical flow [7,8], temporal differencing [9], mixture of Gaussians [10], and point detection or changes [11], just to mention a few. Significant progresses have been achieved, however, important improvements are still possible in many cases, due to a poor image acquisition, variations in the lighting conditions at the scene, shadows projected by foreground objects that could be detected as real objects, multiple objects moving at the scene for both long and short periods and perhaps also occlusions [12]. Object detection tasks increase complexity when carried out on video sequences. Many works have been published for object detection on video sequences acquired by static [13] and non-static cameras [14]. Moving camera segmentation is much more challenging than static camera segmentation because two issues should simultaneously be addressed: background motion and the motion of each foreground-moving object. We refer the reader to [15] for a survey on moving object segmentation methods.

2.2. Agriculture applications

On the other hand, autonomous navigation systems for mobile robots are making an increasing presence in agriculture applications. Image processing techniques have been widely used with discussion to solve problems focused on: identification of weed species [16], mature wheat detection [17], weed elimination [18], crop row detection [19], pest identification [20], estimation of plants nitrogen content [21], and many others applications. Li et al. [22] provided a research review (from 1960s till 2009) on guidance systems and technologies in agricultural vehicles for commercialization of the guidance system. The rapid growth in computing technologies has provided important progress in the field of agricultural autonomous robot guidance systems [23,24]. New technologies have been introduced in recent years to increase the effectiveness of the navigation systems in agriculture applications [25]. These proposals are summarized in Shalal et al. [26], Mousazadeh [27], and Bochtis et al. [28].

2.3. Obstacle avoidance in agricultural environments

A short summary of recent works in precision agriculture applications for obstacle detection task is now included.

Reina and Milella [29] present a self-learning framework for scene segmentation by an autonomous agricultural tractor with

trinocular stereo-vision. The segmentation is based on a self-learning classifier assuming that the vehicle starts its operation from an area free of obstacles. Scenarios analysed include static and non-static obstacles having an average classification rate of 91.0%. In contrast to their work, our methodology does not require a training process; obstacles might appear on scene from the beginning.

Biber et al. [30] incorporated a LIDAR (Light Detection And Ranging) sensor for range finding in their navigation system to detect unexpected objects. Each tractor in the fleet of RHEA was equipped with a LIDAR system to detect, localize, and classify trees with the aim of safe navigation with favourable assessment [31]. The main advantage of LIDAR sensors is that they are not influenced by sunlight with high infra-red radiation content, which is not fully achieved with Kinect sensors. Despite the effectiveness of these LIDAR-based systems, different researches have demonstrated the outperformance of combined LIDAR and vision systems particularly in dynamic outdoor scenarios [32–34].

In [35], authors develop an obstacle detection system using stereo matching to generate a point cloud at regions in the image. Ground plane detection is used to determine which points in the point cloud are obstacles. The detector operates on colour; when the colour of the obstacle becomes similar to that of the terrain the obstacle becomes more difficult to detect. Our proposal is capable of working despite this drawback; we add texture information in order to capture obstacles with similar colour to that of the agricultural environment. Texture is also considered in [36] where the information provided by a Kinect with the LBP (Local Binary Pattern) code is used as a feature to determine if a given pixel is considered as an obstacle or discarded [37]. The algorithm requires a training process to achieve the classifier texture; our method does not require any training process.

Talukder et al. [38] propose a method based on the height difference and slope between three-dimensional points. Despite having been used successfully, the method has a high computational cost. In 2013, Mendes et al. [39] designed a GPU-based parallel version of the obstacle detection proposed in [38] with the aim of improving the efficiency time. The system performance was conducted using point clouds from a stereo camera and an RGB-D sensor. Similar to Mendes, Wei et al. [40] developed a stereo-based system for people detection with the aim of safe navigation in agricultural environments. An obstacle detection system was recently proposed in [41]. 3D points are sampled into a 2D grid, the relations to the neighbourhood are examined and the obstacles are grouped. After, a terrain abstraction is generated. Unfortunately, given the architecture of the fleet in the REA project we do not have additional data; our information is limited to the video sequences acquired with a single camera.

Chein et al. [42] presented an algorithm for localization and mapping based on the detection of olive stems. These are detected by means of both a range sensor laser and a monocular vision system. Authors do not consider obstacle detection as a relevant issue. The methodology presented on this paper is capable of detecting stems; therefore, our method can also be used for path localization and path mapping tasks. Bayar et al. [43] recently introduced a new method for increasing the trajectory tracking performance for autonomous orchard vehicles. In order to detect obstacles and trees they use a laser range finder. They have good performance to detect trees trunks; unfortunately their results do not include obstacle detection analysis.

Gázquez et al. [3] have proposed solutions for humans and robots, working together in agricultural environments. The system is a hybrid between autonomous and classical tele-controlled vehicles. It gives these vehicles some micro programmed intelligence against the unforeseen, and requires a moderate supervision over the robot by the operator. The automaton was programmed to stop when the sensors detect a danger of collision. In contrast with this

proposal, we focus our work on developing an automatic process for obstacle detection based on the analysis of image sequences as expressed above.

A robust algorithm that detects a standardized object has been published in [44]. Authors report a precision of 99.9% in crop rows and 90.8% in grass mowing. The algorithm is based on an existing deep convolutional neural net. However, it is unable to detect people or any different element such as the standardized object. The accuracy of our approach for obstacle detection is not as high as the one reported in [44], but instead, a major advantage is that we do not need any training process, while we also are able to detect elements such as people, trees or vehicles, among others.

As noted throughout this overview, despite the progress reported in the state-of-the-art, there is still work to be done to bring autonomous vehicles particularly in agricultural environments, which is a motivational element of our research.

3. General SDC methodology

The aim of this paper is static/dynamic obstacle detection in agricultural videos; our SDC method involves a sequence of steps where spatial and temporal information is joined to achieve our goal. The video data is processed according to the scheme in Fig. 2 for making a decision at each instant of time during the tractor navigation. The inputs to the system are two consecutive frames, the current frame $I(x, y, z)^t$ and the frame at the previous time $I(x, y, z)^{(t-1)}$; t is the frame index, (x, y) the pixels coordinates, and z represents a specific colour channel, according to the colour space chosen. For simplicity from now on: I^t and $I^{(t-1)}$ are used to refer to the current and the previous frame.

As one can see from Fig. 2, the system involves two main modules where the analysis is carried out at pixel level; the first module is based on the spatial analysis where an image segmentation approach is made for obstacle detection in the current frame, I^t . The segmentation is performed in the LAB colour space based on foreground segmentation; then small spurious areas are removed using morphological operations. A temporal analysis is accomplished in the second module to detect the most prominent changes in the current frame applying a temporal differentiation between I^t and $I^{(t-1)}$. Spatial and temporal information are conveniently combined during the process, obtaining as a final result the static and non-static obstacles in I^t . For a better understanding of the methodology, this section is divided into two main parts; first, obstacle segmentation is explained followed by a motion detection description.

3.1. Spatial analysis

Agricultural outdoor images are complex to analyse mainly because they are captured under different illumination conditions (dawn, sunset, poor or low illumination), most times with high illumination variability during the same working day because of cloud displacements causing important changes in the resulting images. The obstacle segmentation method must be robust enough to deal with these adverse situations. Under these considerations, we studied different colour spaces and found that $L^*a^*b^*$ space is the most appropriate to study agriculture images. The CIELAB colour space was developed in order to obtain an easy-to-calculate colour measurement in accordance with the Munsell colour order system [45]. Furthermore, $L^*a^*b^*$ space is useful for comparing similar colours, which is an included event in our case study.

In cornfield images, during the activities oriented to weed removal, the common elements are green plants (weeds and crops) and soil containing spectral components with high green and red content respectively. Because of this, the majority of obstacles in the cornfield are detected from b^* channel as it represents the

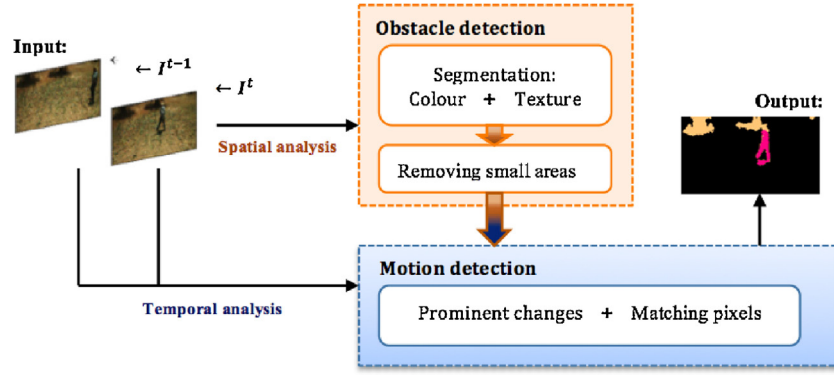


Fig. 2. System illustration for static/dynamic obstacle detection in agricultural videos. The output contains static (brown light colour) and dynamic (pink colour) obstacles detected from input image I^t . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mapping along blue and yellow axis, i.e. colours with low probability of occurrence in corn fields containing the natural elements (plants and soil); however, for this reason obstacles containing a high red and/or white component are not detected from the b^* channel. White and red elements need to be directly identified from L^* and a^* channels respectively. Furthermore, in agricultural environments it is quite common for people to wear clothes with similar colours to the ones existing in the corn fields, i.e. with green and red-brown tones, leading to severe failures during the object detection process based exclusively on spectral information. Fig. 3 shows an example of this failure. The binary image Fig. 3b is the result of the colour segmentation process applied over the image, Fig. 3a. As can be seen, the person on the left of the image is not detected due to their clothes having similar colours to the soil; this problem was also reported in [35]. This is why texture information, Fig. 3c, is required in order to improve object detection as shown in Fig. 3d.

The full scheme of our obstacle segmentation algorithm is portrayed in Fig. 4, where a colour image is split into two complementary sets of pixels: the background, containing pixels from green plants and soil, and the foreground, which contains the remaining pixels. From top to bottom, white (I_{white}), red (I_{red}) and foreground (I_{obs}) detected pixels from L^* , a^* and b^* channels respectively, followed by the texturized image; from which pixels having different texture than soil and green plants are extracted (I_{btex}). The information coming from I_{white} , I_{red} , I_{obs} , and I_{btex} is concatenated in a single image. The resulting binary image contains small and spurious areas; they are removed to retain the main areas, which are potential objects. The final binary image I_{bin} can be appreciated at the bottom in Fig. 4. The complete process to obtain these intermediate images is outlined in the steps 1a, 2a and 3a.

Before proceeding, it is necessary to define the statistics measured used to compute the threshold values τ_1 and τ_2 for segmenting an image of size $H \times W$. Let $I(x, y)$ be the intensity of

pixel (x, y) . The mathematical definition of the mean (μ), standard deviation (σ) and skewness (s) are given by expressions (1)–(3) respectively. The threshold values τ_1 and τ_2 are calculated by way of Eqs. (4) and (5), where n is the number of standard deviations considered.

$$\mu = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W I(i, j) \quad (1)$$

$$\sigma = \sqrt{\frac{1}{H \times W - 1} \sum_{i=1}^H \sum_{j=1}^W (I(i, j) - \mu)^2} \quad (2)$$

$$s = \frac{(1/H \times W) \sum_{i=1}^H \sum_{j=1}^W (I(i, j) - \mu)^3}{\left[(1/H \times W - 1) \sum_{i=1}^H \sum_{j=1}^W (I(i, j) - \mu)^2 \right]^{3/2}} \quad (3)$$

$$\tau^1 = \mu + n \times \sigma \quad (4)$$

$$\tau^2 = \mu - n \times \sigma \quad (5)$$

Continuing with the methodology description. Given the current frame $I(x, y, z)^t$ in RGB colour space follow steps 1a, 2a and 3a to achieve the obstacle segmentation desired.



Fig. 3. (a) Colour image. (b) Segmentation based on $L^*a^*b^*$ colour space. (c) Texture extraction. (d) Final segmentation combining (b) and (c) (I_{bin}^*).

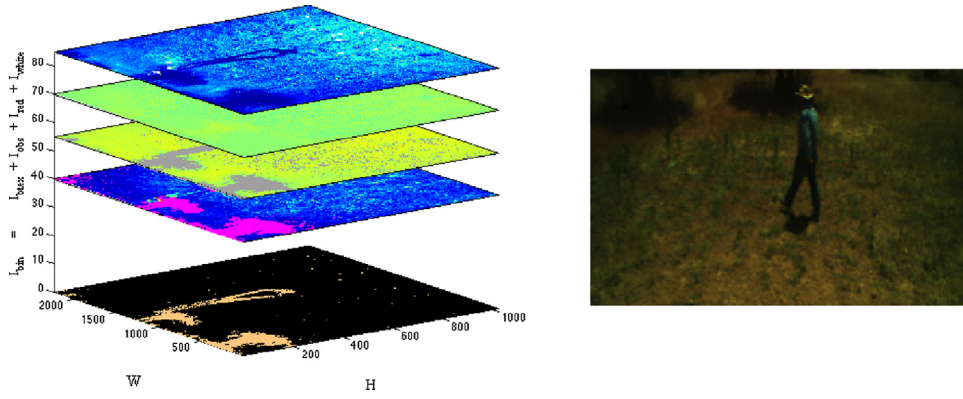


Fig. 4. Obstacle detection process applied to the colour image (H-by-W size) on the right.

Algorithm 1. Image segmentation. Foreground extraction in agricultural images.

Input: I_b, b^* – channel from $L^*a^*b^*$ colour space.
Output: I_{obs} , binary image (1: foreground, 0: background).

```

1  From  $I_b$  compute: the
   frequency histogram ( $h_b$ ), the
   skewness  $s_b$  (Eq. (3)) and the
   threshold  $\tau_b^1$  (Eq. (4)) values;
   Compute the critical points
   from  $h_b$ . Applied finite
   differences method [46] was
   used here;
2
3   $thl$  = Global maximum criticalpoint;
4  if  $s_b \leq 0$  then
5    return  $T = thl$ ;
6  else
7
8     $thl_1$  = Minimal critical point before
   the global maximum critical point;
   return  $T = \min(\tau_b^1, thl, thl_1)$ ;
9  end
10 Threshold image as:  $I_{obs}(x, y) =$ 
     $\begin{cases} 1 & \text{if } I_b(x, y) < T \\ 0 & \text{otherwise} \end{cases}$ 

```

Step 1a Spectral segmentation: First, change RGB image into $L^*a^*b^*$ colour space to obtain L^* (I_L), a^* (I_a) and b^* (I_b) channels respectively. Apply Algorithm 1 over I_b to get a binary image I_{obs} where, pixels with value “1” represent obstacles and pixels with value “0” represent background. As we mentioned before, b^* channel is related to the blue-yellow axis and for this reason Algorithm 1 does not capture objects with high red spectral components. To emphasize objects composed of red pixels we make use of I_a . In a^* -channel, green colour is in the negative axis while red colour is in the positive axis. Red pixels are filtered with the expression: $I_{red}(x, y) = 1$ if $I_a(x, y) > \tau_a^1$, 0 otherwise. Note that the threshold value τ_a^1 is computed from the intensity values in I_a using Eq. (4). In order to capture white elements, a similar process with red detection is made. From luminosity channel next operation is applied: $I_{white}(x, y) = 1$ if $I_L(x, y) > \tau_L^1$, 0 otherwise.

Step 2a Texture extraction: The texture image of $I(x, y, z)^f$ is computed from the grayscale image $I_{grey}(x, y)$. For each intensity value in I_{grey} ; the difference between the maximum and minimum grey level over a ω -by- ω neighborhood $\Omega(x, y)$ centred on the interest pixel (x, y) is the texture value of the pixel, Eq. (6). From the texture image I_{tex} , the segmentation is carried out as: $I_{btex}(x, y) = 1$ if $I_{tex}(x, y) < \tau_{tex}^2$, 0 otherwise, the threshold value τ_{tex}^2 is computed with Eq. (5).

$$I_{tex}(x, y) = \max(I_{grey}(x_i, y_i)) - \min(I_{grey}(x_i, y_i)), (x_i, y_i) \in \Omega(x, y) \quad (6)$$

Information coming from spectral segmentation and texture extraction is combined to achieve the full obstacle segmentation I_{bin}^* , Eq. (7).

$$I_{bin}^*(x, y) = (I_{red}(x, y) + I_{white}(x, y) + I_{obs}(x, y) + I_{btex}(x, y)) > 0 \quad (7)$$

Step 3a Removal of small areas: Once I_{bin}^* is obtained containing potential obstacles, small or spurious regions are suppressed under the assumption that they are irrelevant from the point of view of safe autonomous navigation. To perform this task a morphological opening [47] is carried out in I_{bin}^* having as a result the binary image I_{bin} . This contains the relevant obstacles on the current frame $I(x, y, z)^f$. Opening, as it is known is the combination of two fundamental morphological operations, an erosion followed by a dilation. This allows removing small spurious binary areas, isolated pixels or pixels on object boundaries. The number of pixels removed from the objects in an image depends on the size and shape of the structuring element used to process the image. In this work, the structuring element chosen for both operations was a circle with different sizes, as discussed below.

Due to the camera imaging perspective projection Fig. 1a, the same object in a frame located at different spatial positions in the 3D scene is mapped onto the image with different sizes, measured in pixels; its size is small at the top and big at the bottom of the image. For this reason, erosion is not uniformly carried out on the binary image. Circles with different radio lengths were used in the erosion, taking the image height as a reference. Four areas are identified, from the top to the bottom; the erosion was performed using circles of radio 1, 2, 3 and 4 pixels respectively, which are sufficient according to the camera vision system setup. Dilation was uniformly performed throughout the whole image using a circle of radio 1. Fig. 5 shows the result after removing small areas in the segmented image in Fig. 3d.

From now on, we are focus on identifying which elements on I_{bin} are stationary or moving. To discriminate among static and non-static pixels temporal information is required.

3.2. Temporal analysis

There are many existing schemes in literature for motion detection, they were described in Section 2. The simplest and most popular approach involves two consecutive frames. This is the base of our proposal for discriminating among static and non-static obstacles on cornfield videos. We consider the information provided by the previous frame $I^{(t-1)}$ to detect moving pixels in I^t using as reference the obstacle segmentation mask I_{bin} . Now, description about how discrimination among static and non-static motion is carried out.



Fig. 5. Left: I_{bin}^* , see Fig. 3d. Right: results after removing small areas, I_{bin} .

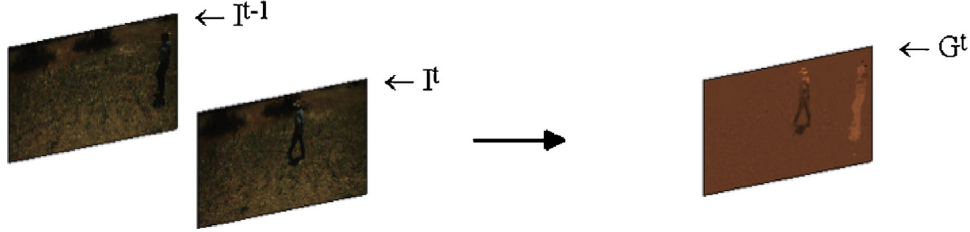


Fig. 6. Left: Two consecutive frames; I^t and I^{t-1} . Right: The temporal gradient G^t computed from I^t and I^{t-1} .

Step 1b Prominent motion detection: The temporal gradient is extracted using the first order differential approximation: $G^t = I^t - I^{(t-1)}$. The prominent changes in G^t are obtained by thresholding. From G^t , two threshold values, τ_G^1 and τ_G^2 , are computed with the expressions (4) and (5) respectively. These values are used to get two binary images d_1^t and d_2^t by Eqs. (8) and (9) respectively.

$$d(x, y)_1^t = \begin{cases} 1 & \text{if } G^t(x, y) > \tau_G^1 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

$$d(x, y)_2^t = \begin{cases} 1 & \text{if } G^t(x, y) < \tau_G^2 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

In Fig. 6, the temporal gradient from two consecutive frames is portrayed, their resulting binary images d_1^t and d_2^t are displayed in Fig. 7. As shown in the images, pixels with major changes at the scene can be detected.

Step 2b Matching process: To determine which of the two masks is the most convenient, i.e. the one containing the maximum movement information in the current frame, we proceed as follows: d_1^t and d_2^t are independently overlaid on top of the obstacle segmentation I_{bin} . From each overlapping the number of matching pixels are counted using expressions (10) and (11) having as a result M_1 and M_2 . The mask with a major number of matching pixels $\max(M_1, M_2)$ is chosen as the moving mask for the current frame I^t .

$$M_1(I_{bin}, d_1^t) = \sum_{i=1}^H \sum_{j=1}^W \{d(i, j)_1^t \mid d(i, j)_1^t + I_{bin}(i, j) > 1\} \quad (10)$$

$$M_2(I_{bin}, d_2^t) = \sum_{i=1}^H \sum_{j=1}^W \{d(i, j)_2^t \mid d(i, j)_2^t + I_{bin}(i, j) > 1\} \quad (11)$$

Continuing with the example in Fig. 7, d_2^t contains the greatest number of matched pixels with I_{bin} ; $M_1 < M_2$. This can be seen in Fig. 8, I_{bin} is on the base of the two graphics, over it d_1^t (on the left) and d_2^t (on the right). Pixel pairs matching are highlighted in pink colour at the base and on top of the graphics. To conclude, also masks on the base of the graphics show static (light brown colour) and non-static (pink colour) pixels.

4. Performance evaluation

The experiments reported in this work were performed with videos acquired with a SVS4050CFLGEA sensor model (SVS-VISTEK, 2013) during April/May/June 2012–2014 in a 1.7 ha experimental field of corn in La Poveda Research Station, Arganda del Rey, Madrid. Videos were taken with a frame rate of 2 fps. All acquisitions were spaced by five/six day periods, that is, they were obtained under different illumination conditions and different plant growth states. The CCD is a Kodak KAI 04050M/C sensor with a Bayer colour filter with GR pattern; resolution of 2336×1752 pixels and 5.5×5.5 μm pixel-size. A set of 110 videos of different length, the longest with 2099 frames while the shortest with 20 frames (about 15,000 frames), compose our dataset including videos during the final demonstration of the RHEA project in May 2014. There are four types of events that can be observed in our dataset, they are described in Table 1.

It is worth noting that in scenarios 1 and 3, although the tractor is stopped, small movements are detected, derived from the motor vibrations when it is turned on. In order to provide a quantitative evaluation of our proposals, statistical measures described in Table 2 are used to measure the performance of the algorithms. The constants; TP , FN and FP are the true positive, false negative and false positive outcomes.

There are two parameters involved in the whole process; the number of standard deviations considered in Eqs. (4) and (5), and the window size related to the texture extraction, Eq. (6). In the first case, n is set as 2. Experimental results have shown a good efficiency for all experiments carried out. As regards the window size ω required to define the neighbourhood area for texture extraction, this value must be conveniently fixed. If the block size is too large, the probability that the region contains different textures is high. On the other hand, if the block size is too small, the capture

Table 2
Performance evaluation metrics.

Name	Abbreviation	Equation
Sensitivity or true positive rate	TPR	$TP/(TP + FN)$
Precision or positive predictive value	PPV	$TP/(TP + FP)$
False negative rate	FNR	$FN/(FN + TP)$
False discovery rate	FDR	$1 - PPV$

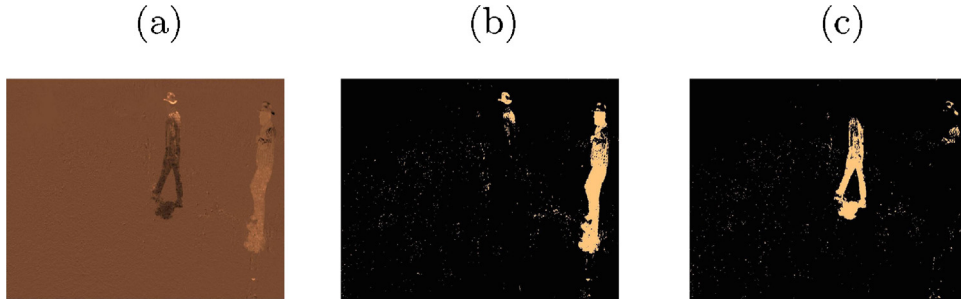


Fig. 7. (a) Temporal gradient G^t , Fig. 6. (b) and (c) Binary images d_1^t and d_2^t .

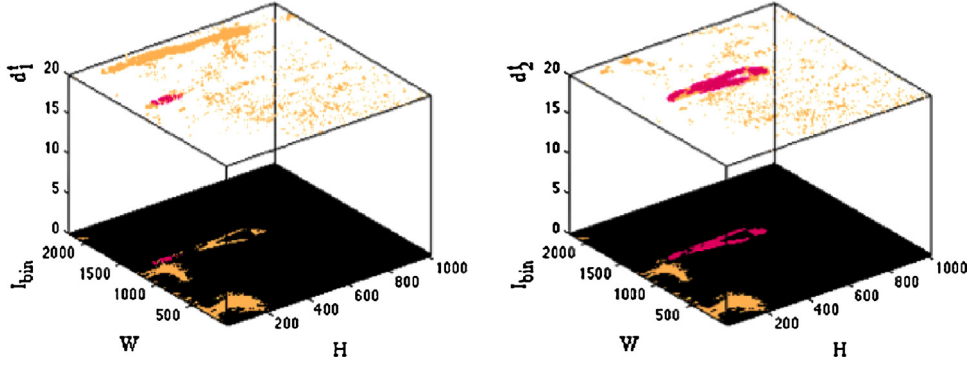


Fig. 8. Matching pixels for motion detection: Left: I_{bin} on the base, on top in light brown colour and transparent background d_1^t (Fig. 7b). Right: I_{bin} on the base and d_2^t (Fig. 7c) at the top. Pixel pairs matching are highlighted in pink colour. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of sufficient texture information becomes difficult. It is not easy to define an optimum block size. We tested different sizes with $\omega = \{3, 5, 7, 9, 11\}$ where a trade-off between the image resolution and the block size must be established. After different trial and error experiments we have verified that in our camera system setup the best performance was achieved with $\omega = 9$.

From now on, quantitative and qualitative results are given. Firstly, obstacle segmentation outcomes are provided. Secondly, motion detection as well as the evaluation of the discrimination among static and non-static obstacles under all scenarios described in Table 1 is presented. In both cases, performance is compared against methodologies well known in the state-of-the-art.

4.1. Obstacle segmentation

To measure segmentation quality, the success rate (SR) is obtained as follows. Given a ground truth bounding box (TB) and the segmentation result (S) achieved with the process described in Section 3.1, SR is computed as $area(TB \cap S)/area(TB)$. In Fig. 9 the process is illustrated using a single image where two obstacles are present on the frame. In Fig. 9b the ground truth-bounding box for the obstacles in Fig. 9a while in Fig. 9c the segmentation obtained

using our proposal. The SR achieved for obstacles 1 and 2 were 97.65% and 97.55% respectively.

Now, with respect to the evaluation for each case in Table 1, we made the next process. From the video dataset 80 frames were randomly chosen. This selection includes 20 images per scenario. The regions considered as obstacles were manually touched up, pixel by pixel, to delimit and fill gaps at each obstacle. A bounding box is drawn for each obstacle representing the TB. The total number of true bounding boxes obtained was: 31, 36, 29 and 51 for scenario 1, 2, 3, and 4, respectively (second column of Table 3). After that, the 80 images were processed by means of the algorithm described in Section 3.1 to obtain the corresponding obstacle segmentation (S). Results were compared with the truth bounding boxes to get the SR for each obstacle. Finally, the average success rate for each scenario is presented in the third column of Table 3. Similar to the process described before, we obtain the average SR results using: Otsu [48] and Fuzzy c-means (FCM) [49] segmentation algorithms. Results are visible in the fourth and fifth columns in the same table.

Now, for a broader evaluation of how the obstacle segmentation process is working, the number of images tested was increased. The output of a set of 300 frames was visually inspected to determine when a success occurs (TP- obstacle detected) or when a

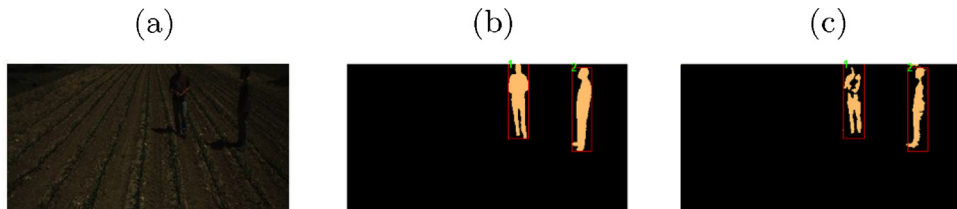


Fig. 9. (a) Colour image. (b) Obstacles on the left image manually segmented. They are enclosed in red boxes and represent the ground-truth (TB). (c) Obstacles detected with our segmentation algorithm (S); the SR values are 97.65% and 97.55% for obstacle 1 and 2 respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3

Average success rates (%) achieved with different segmentation processes.

Case	# Obstacles	Our segmentation approach	Otsu	FCM
Scenario 1	31	74.96	16.39	66.14
Scenario 2	36	85.93	20.31	66.12
Scenario 3	29	85.61	24.39	65.72
Scenario 4	51	86.38	19.70	63.14

Table 4

Evaluation of the metrics in Table 2 for the obstacle detection algorithm, Section 3.1.

Metric	Evaluation (%)
TPR	93.88
PPV	83.36
FNR	6.12
FDR	16.64

failure takes place (FN – obstacle non-detected, FP – non-existent obstacle detection). The numerical results of the evaluation of the expressions in Table 2 are shown in Table 4. The obstacle detection with the process described in Section 3.1 shows good performance. The proportion of obstacles correctly identified is 93.88% while the percentage of positive predictive value is 83.36%. Accuracy rate is similar to the results reported by Reina and Milella [29].

Other obstacle detection results are given in Fig. 10, four consecutive frames from a video sequence are presented in the first row. Trees detected in each frame using our algorithm are in the second row; now we have outlined the contours defining the objects instead of the binary results. Stems can be used for path planning of the autonomous vehicle, as suggested in [3,42] and [43], where stems are detected as part of the navigation system.

Up to this point we have presented the results related to object obstacles detection based on spatial analysis. Motion detection results are thoroughly discussed below.

4.2. Motion detection

We compared our proposal against four well-known methodologies reported in literature: optical flow estimation with LucasKanade (Of-lk) and Tikhonov regularization (Of-tr)[50], temporal differencing (Td) [9], and mixture of Gaussians (MoG) [10]. Details concerning these four methods can be found in cited papers. For optical flow results, the pixels with major optical flow values were filtered by a threshold value of 0.5 as suggested in [50]. All methods, including our proposal, were tested with consecutive frames in the four scenarios. Some visual results are displayed in Table 5; the consecutive frames can be seen in the first and second rows in Table 7. In Table 5 methods appear in the rows while the scenarios are displayed in the columns. The pink colour in images

represents motion detection. The ideal outcome in this table should be when no pink marks appear in scenarios 1 and 2, in contrast to scenarios 3 and 4, pink colour must only be applied on the moving obstacles. The best outcome in Table 5 occurs when the tractor is stopped, scenario 1 and 3. In these scenarios, the performance of Of-tr, Td and SDC is acceptable, while Of-lk and MoG fail because of motion is detected on static regions. The poor results, including our methodology, happen when the tractor is operating. This failure is clearly remarkable in scenario 2 where motion is wrongly detected on regions where nothing happens. Here, the best outcome is achieved with Of-tr. The weak point of this approach occurs in non-static obstacles as can be seen in scenario 4 (second row fourth column). The method Of-tr has problems to remove motion that belongs to the previous frame. In this regard, SDC works well by the fact that the motion mask is selected using the obstacle segmentation in the current frame as reference. This information help us to remove motion coming from the previous frame. In the opposite case, our proposal still has a poor level of performance in scenario 2.

Table 5

Motion detection in consecutive frames for scenarios in Table 1. The consecutive frames can be seen in the first and second rows in Table 7.

	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Of-lk				
Of-tr				
Td				
MoG				
SDC				

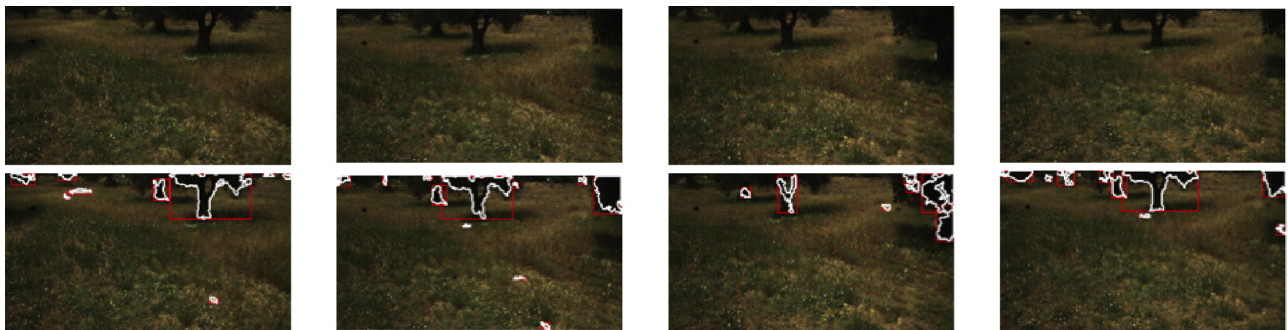
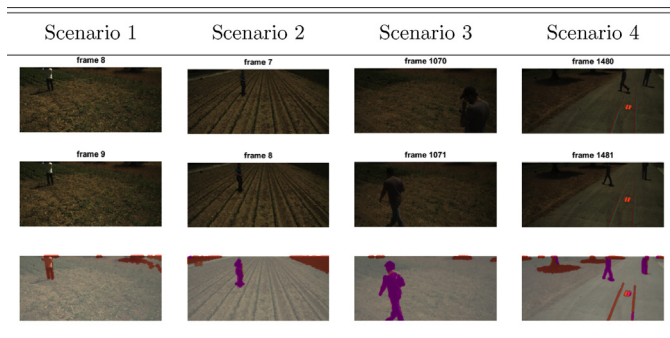


Fig. 10. Consecutive frames. Trees trunks detected are delimited by using white lines and enclosed in red boxes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6
Percentage of success for motion detection based on the methodologies in Table 5.

	OF-lk	OF-tr	TD	MoG	SDC
Scenario 1	55.14	79.83	88.88	54.15	98.78
Scenario 2	66.08	86.52	54.05	22.88	61.11
Scenario 3	61.12	78.68	76.54	30.22	89.88
Scenario 4	59.68	32.41	28.77	39.92	75.20
Average success percentage	60.51	69.36	62.06	36.79	81.24

Table 7
First and second row: consecutive frames under the different scenarios. Last row: obstacle detection (in light brown) and motion detection (pink colour) obtained with SDC algorithm.



The qualitative results in Table 6 reinforce the foregoing. In this Table, the qualitative performance evaluation of motion detection is obtained by counting the number of successes achieved in the process to identify static and non-static obstacles in the agricultural video sequences. For scenarios 1 and 2 an event is considered a success if no motion is detected. In contrast, if motion is detected then the event is considered failed. In scenarios 3 and 4, the event is a success in any of these cases: motion is detected on static elements or motion is detected on moving obstacles. A fault occurs when motion is detected in static elements or motion is not detected in

moving obstacles. The quantitative performance is expressed in terms of percentage of success. From Table 6 also OF-tr and SDC have the best average successes percentage, the first for scenario 2 and the second in the remaining scenarios. In this table it is important to highlight that the accuracy in scenario 4 using OF-tr decreases significantly because the number of non-static obstacles is larger than static obstacles in videos for scenario 4.

It should be noted that in the fifth row in Table 5, results with our SDC method, only motion detection is displayed. Results including the obstacle segmentation are given in Table 7, where obstacle segmentation is represented with light brown colour and motion detection is highlighted with pink colour. Motion is the same as the last row in Table 5. Also in this table, the two consecutive frames used to get results in Table 5 are given in the first and second row.

The results hereafter refer to the performance assessments of the discrimination between static and non-static elements; metrics in Table 2 are used for this task. Definitions for *TP*, *FN* and *FP*, are different for static and dynamic obstacles, they are given in Table 8

After applying our SDC algorithm to the videos database, results were visually inspected to determine when the algorithm performs successfully or fails. The results are reported in Table 9. As earlier noted, the algorithm shows good performance to detect among static and non-static obstacles when the tractor is stationary (scenario 1 and 3), however, performance is poor when the tractor navigates (scenarios 2 and 4). Still, it is difficult to identify if the

Table 8
Definitions for metrics in Table 2. They are used to evaluate the performance of our SDC algorithm to discriminate among static and non-static elements.

True state	Apply	Terminology	Definition
Static obstacle	Scenarios 1 and 2	True positive (<i>TP</i>)	No detected motion in the obstacles, they are correctly static
		False positive (<i>FP</i>)	No detected static obstacles on the frame
		False negative (<i>FN</i>)	Detected motion in the obstacles given that obstacles are static
Dynamic obstacle	Scenarios 3 and 4	True positive (<i>TP</i>)	Detected motion in the obstacles given that they are moving
		False positive (<i>FP</i>)	No detected moving obstacles on the frame
		False negative (<i>FN</i>)	No detected motion in the obstacles given that they are moving

Table 9
Average performance evaluation of the SDC algorithm to discriminate among static and non-static obstacles.

Metric(%)	Scenario 1	Scenario 2	Scenario 3	Scenario 4
TPR	93.83	34.29	91.89	63.03
PPV	96.75	39.85	82.88	80.33
FNR	6.16	19.32	8.10	36.97
FDR	3.25	13.76	17.11	19.67

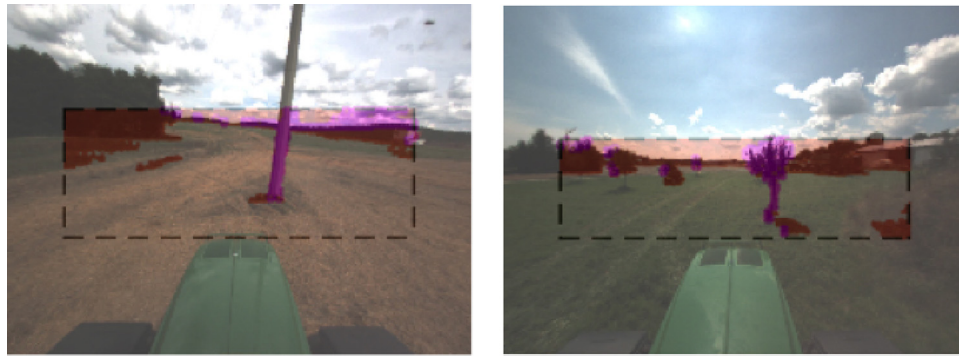


Fig. 11. Obstacle detection in pair of images acquired by a stereo-vision device. SDC process was applied to the region enclosed by dotted black lines. Obstacles detected are in brown, while motion detection is in pink. Images courtesy of P. Fleischmann [41]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 10

Different obstacles detected with SDC, our detector successfully detects people and obstacles in different sample fields. Images in the third and fourth columns were respectively acquired from [52] (courtesy of A. Ribeiro and co-authors) and from [51] (courtesy of Arno Ruckelshausen and co-authors).

Scenario 1	Scenario 2	Scenario 3	Scenario 4
frame 4	frame 103	frame 152.001	frame 94.04
frame 5	frame 104	frame 153.001	frame 95.04

motion flow comes from the tractor motion or from dynamic obstacles.

Trying to increase the comparative results we tested our algorithm on videos from others projects [51,52], results are visible in Table 10. Images displayed in column 3 are obtained and used in [52] for path planning. Images in the fourth column are acquired from [51]; the video shows manoeuvres and applications for research robot platform “BoniRob” [30]. These images are courtesy of authors in these references.

Images in Fig. 11 come from a stereo-vision device; they are tested and included in the experimental results section in [41]. The SDC output is visually similar to those exposed in [41]. Our results are also comparable to those presented by Ross et al [35] where the obstacle was a person. As can be seen, our algorithm is capable of detecting obstacles in different crop fields and under different light conditions. The combination of temporal and spatial information has been shown enough for obstacle detection, mainly for people appearing in the crop field.

Another important consideration is that our system can manage favourably the challenge of working under high variability of lighting conditions. Nevertheless, shadows are still a problem, causing failures during object detection. Table 11 shows some examples where our approach fails, shadows are detected as obstacles. The behaviour explained with these examples is also seen for the set of images analysed. Wrong detection of shadows as obstacles is included in the statistical results in Table 9.

To close the section, the results presented were computed using the Image Processing Toolbox MATLAB 2013a for 64 bits under Windows 7 and Intel Core 2 CPU, 3 GHz, 4 GB RAM.

Table 11

Examples of shadows detected as obstacles using SDC algorithm.

Scenario 2	Scenario 2	Scenario 3	Scenario 4

5. Conclusions and future trends

In an attempt to discriminate among static and non-static obstacles we carried forward the work of obstacle detection in RHEA project [31]. Identifying dynamic obstacles, people walking into the crops area, is important for preventing and ensuring the safety of people/animals and also to guarantee the continuity of the process. Our proposal is based on image processing analysis supported by the combination of spectral segmentation and temporal differences. Firstly, obstacle detection is performed in the spatial scale and then, a temporal differencing is carried out for motion detection. This approach addresses the problem of uncertainty in uncontrolled outdoor agricultural and dynamical scenarios involving agricultural machinery. Although the proposed approach could be considered deterministic in nature, the high variability in the illumination conditions, camera vibrations, unknown and unexpected static or dynamic objects introduce the aforementioned uncertainty. Hence, the proposed computer vision approach deals with differences in initial conditions, belonging to a kind of systems with a chaotic deterministic behaviour in the context of applied soft computing in agricultural machinery.

The experimental results show that our method is suitable and effective in the context of our agricultural scenarios. We also show that trees detection is possible from the point of view of image analysing processing, making it a useful contribution for path localization and path mapping tasks. More specifically, another important contribution is that no training process is required. It is entirely based on a combination of digital and image analysis techniques.

Object detection on video sequences is still an open issue demanding special attention in view of the commercialization of these vehicles in the future. We have established an important basis

for continuing towards finding a global solution to the problem of detecting obstacles for safe navigation in autonomous agricultural vehicles. For future work we propose exploring the following possibilities to improve the results hereby presented:

- i Results in Tables 5 and 6 suggest that OF techniques combined with our proposal can be beneficial in order to discriminate between static and non-static obstacles.
- ii The segmentation algorithm described in Section 3.1 has been improved with temporal analysis. However, the results also show that the algorithm is not able to handle shadows, even people shadows. The process still has failures in this context; shadows are difficult to identify, requiring an additional effort in this regard.
- iii Animals, people, cars, fence posts, trees, buildings, and so on, need to be correctly classified in order to gain maneuverability and autonomy for vehicles in the field.
- iv Motion detection can be used for performing a tracking task of dynamic objects at the scene. More specifically, prediction of activities to detect people walking, standing or running in the crop fields can be included to increase safe navigation with autonomy.

Convincing solutions to all of these problems are required to reduce the uncertainty introduced by all factors affecting the system, including the ones proposed for the future. This reduction should lead to the minimization of differences in the initial conditions of the vision system improving its chaotic behaviour in the context of soft computing.

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